Reli

1. Scope & Feed Conditions

First-principles assessment of three hydrocarbon recovery technologies for a 650 kg/hr purge stream at 1.20 bar(a) and 64 °C (5 mol% C_3H_6).

Parameter	Value
Feed mass flow	650.0 kg/hr
Feed pressure	1.20 bar (absolute)
Feed temperature	64.0 °C
N2 mole fraction	95 mol%
HC mole fraction (C3H6)	5 mol%

2. Recovery Efficiency Comparison

The chart below shows the net monomer recovery efficiency for each technology under the stated feed conditions. The dashed red line marks the minimum plant target of 90%.

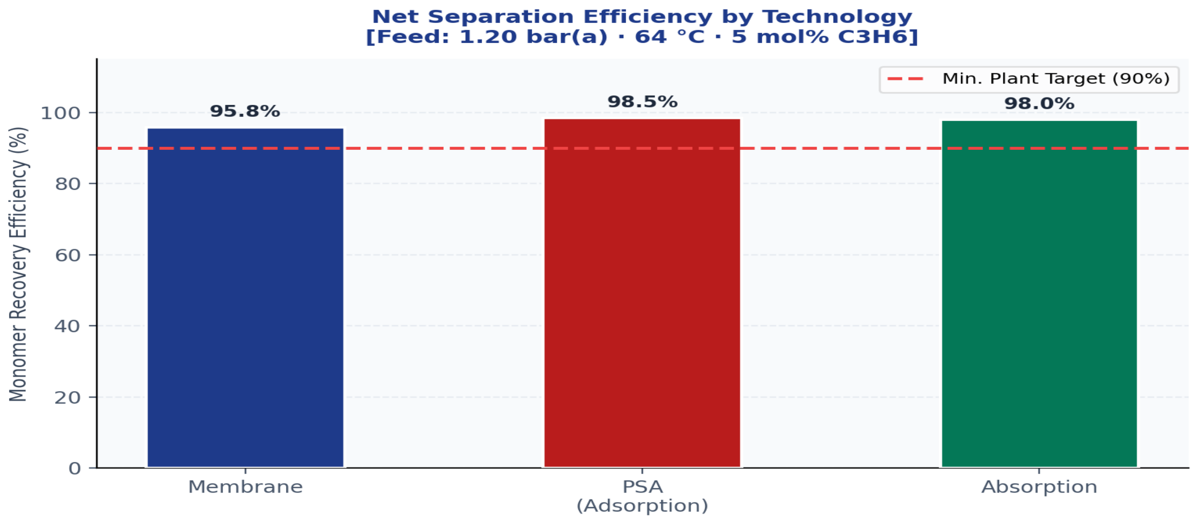

Figure 1 — Net Separation Efficiency (η_{global}) by Technology

3. Capital Expenditure (CAPEX) Comparison

CAPEX estimates are derived from standard catalog equipment using published scaling exponents (cost index year 2026). Membrane cost is dominated by the feed gas booster compressor required to raise the 1.2 bar feed to 16.2 bar permeate differential.

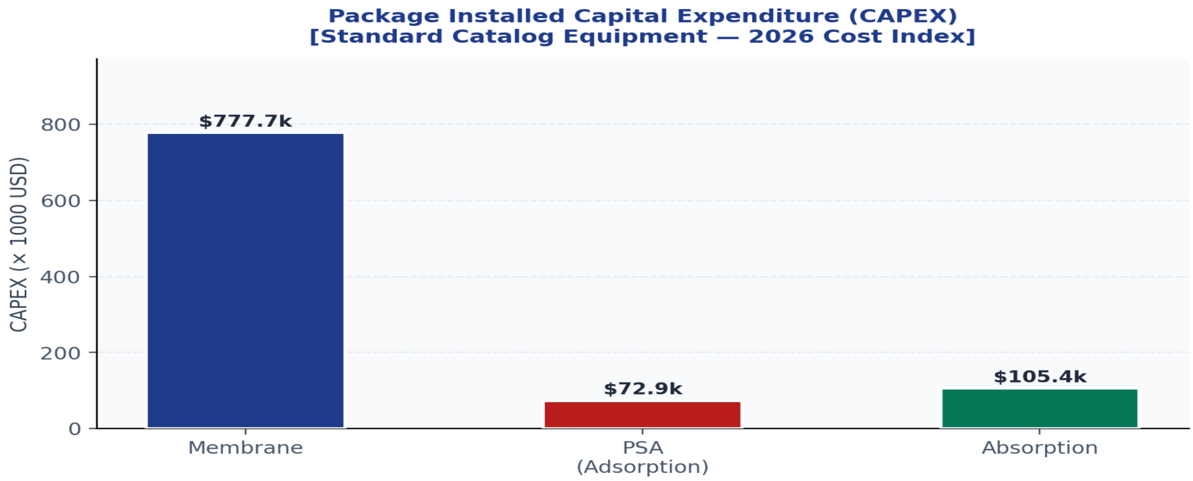

Figure 2 — Installed Package CAPEX (USD). Feed compression excluded for PSA and Absorption.

6.1 Component Breakdown — Hollow-Fiber Polymeric Membrane Skid System

Component	Rating	Max Capacity	Used Capacity	CAPEX (USD)	OPEX ($/hr)
K-301 Feed Gas Booster Compressor	API 619 Screw Machine / 110.0 kW Motor Enclosure	126.5 kW thermal limit	94.6 kW compression work	$172,000	$8.80
M-301 Polyimide Membrane Separator Skids	ASME Rated Bank housing 14 Standard Elements	560.0 m2 active area barrier	537.6 m2 selective flux interface (log-mean delta-P = 0.241 bar)	$605,700	$25.49
TOTAL				**$777,700**	**$34.29**

K-301: *Isentropic Pressure Ratio: 13.5:1 | p_discharge = 16.2 bar (from process_parameters.json)*

M-301: *Calculated Permeate Matrix Gas Recovery: 95.8% | OPEX breakdown: replacement + auxiliaries = $25.49/hr*

Net Recovery Efficiency: 95.8% · Total Package CAPEX: $777,700 · Total OPEX: $34.29/hr

6.2 Component Breakdown — Twin-Bed Cyclic Pressure Swing Adsorption Skid

Component	Rating	Max Capacity	Used Capacity	CAPEX (USD)	OPEX ($/hr)
Y-201 Duplex Guard Cartridge Element	ASME Housing / 0.5 Micron Particulate Strainer	740.3 m3/hr hydraulic ceiling	528.8 m3/hr actual volumetric flow	$18,500	$0.45
V-201 A/B Twin Swing Adsorber Steel Shells	ASME Section VIII / ANSI 150# Rating Vessel Frame	230.7 kg standard loading frame	142.6 kg required sieve mass	$54,400	$3.50
TOTAL				**$72,900**	**$3.95**

Y-201: *99.95% Resin Fines Separation Guard*

V-201: *Langmuir Pore Vapor Equilibrium Affinity: 98.5% | Cycle: 10 min total / feed step 40% (4.0 min)*

Net Recovery Efficiency: 98.5% · Total Package CAPEX: $72,900 · Total OPEX: $3.95/hr

6.3 Component Breakdown — Heavy Hydrocarbon Gas Absorption Recirculation Loop

Component	Rating	Max Capacity	Used Capacity	CAPEX (USD)	OPEX ($/hr)
E-101 Feed Gas Cooler (Shell & Tube)	TEMA R / ANSI 150# — CW Duty Gas Cooler	15.2 m2 surface (25% margin envelope)	12.2 m2 required transfer surface | Duty: 5.7 kW	$19,200	$0.00
T-101 Absorber Tower Packed Shell	ASME Section VIII / ANSI 150# Low Press Casing	1142.2 kg/hr hydraulic ceiling	650.0 kg/hr @ 35 °C (post E-101)	$76,800	$9.57
P-101 A/B Solvent Circulation Pumps	API 610 Centrifugal / Standard 7.5 kW Frame	12.0 GPM impeller cut	9.6 GPM solvent circulation	$9,400	$8.92
TOTAL				**$105,400**	**$18.49**

E-101: *Cooling duty: 64.0 °C → 35.0 °C | delta-T_lm = 10.5 K | NOTE: E-101 is a missing upstream unit — without it the absorber operates at feed temperature (64 °C), raising Henry's constant and requiring ~65× L/G to compensate.*

T-101: *Thermodynamic Recovery Yield: 98.0% | Absorption Factor A = 1.50 | L/G (mass) = 2.7 | C_design = 0.0770 m/s (Mellapak 250Y @ 72% flood)*

P-101: *Solvent Recirculation Loop Containment*

Net Recovery Efficiency: 98.0% · Total Package CAPEX: $105,400 · Total OPEX: $18.49/hr

7. Technology Comparison Matrix

Qualitative and quantitative comparison across key engineering criteria.

Criterion	Membrane	PSA	Absorption
Recovery Efficiency (%)	95.8%	98.5%	98.0%
CAPEX (USD k)	$777.7k	$72.9k	$105.4k
Meets 90% Target	Yes	Yes	Yes
Operability	High	Medium	High
Maintenance	Low	Medium	Medium
Footprint	Medium	Large	Large
Recommended	✓ YES	✓ YES	✓ YES

8. Key Findings & Recommendation

1. Polymeric Membrane achieves 95.8% recovery — meets target; CAPEX $777.7k.
2. Pressure Swing Adsorption achieves 98.5% recovery — meets target; CAPEX $72.9k.
3. Heavy Gas Absorption achieves 98.0% recovery — meets target; CAPEX $105.4k.

ENGINEERING RECOMMENDATION

Pressure Swing Adsorption is the primary recommendation, meeting the recovery target at the lowest CAPEX ($72.9k). Polymeric Membrane, Heavy Gas Absorption also meet the target and remain viable alternatives.